



Interface Design

IDE2IL

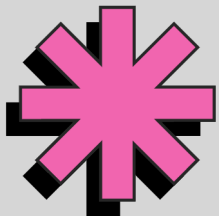


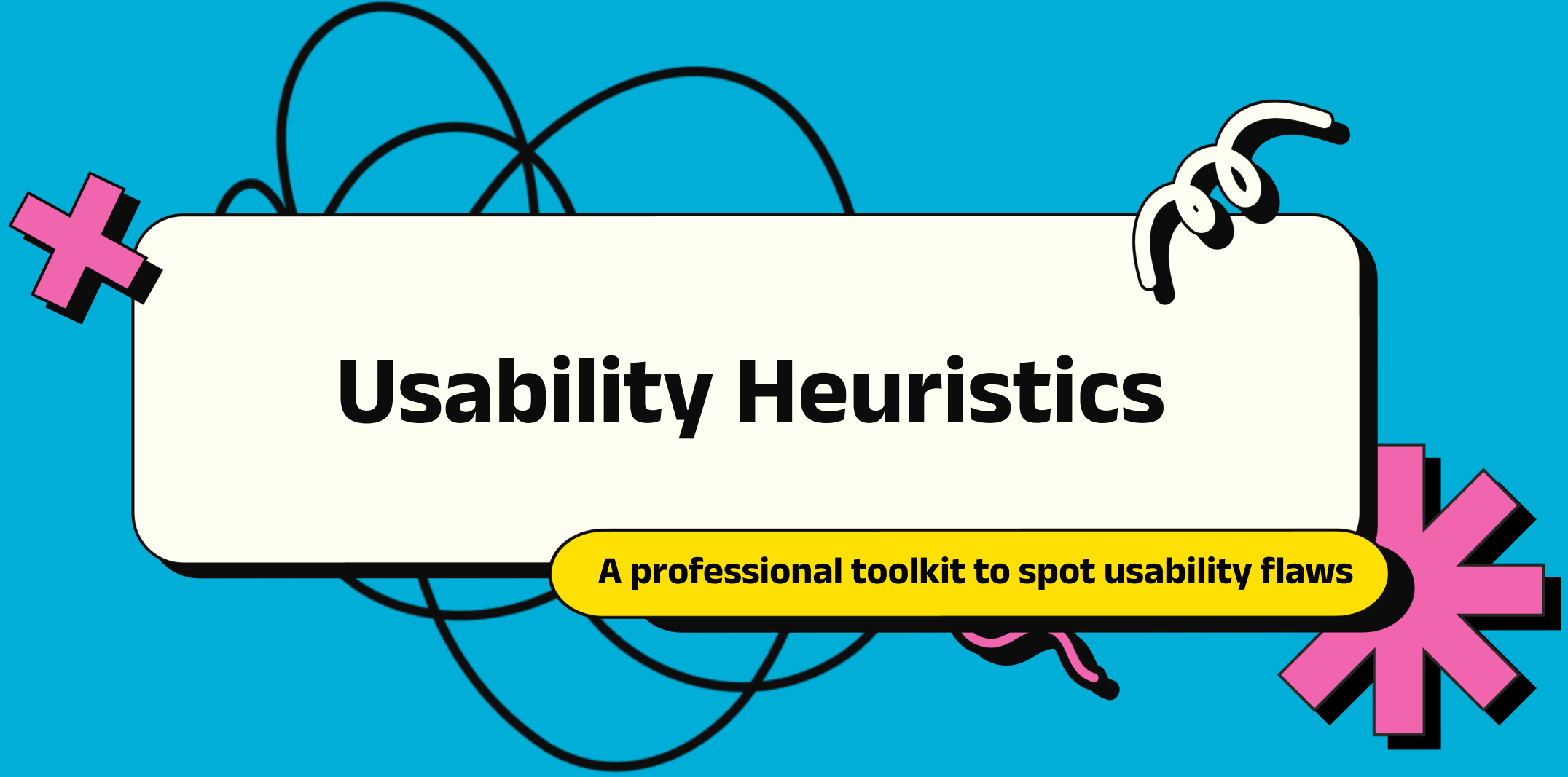
kathrin.probst@fh-ooe.at



TODAY'S GOAL

**DISCOVER THE 10 FUNDAMENTAL
HEURISTICS OF INTERFACE DESIGN.
MASTER THE RULES OF USABILITY
THROUGH HANDS-ON EVALUATION.
GET READY TO BUILD INTERFACES
THAT JUST WORK!**





Usability Heuristics

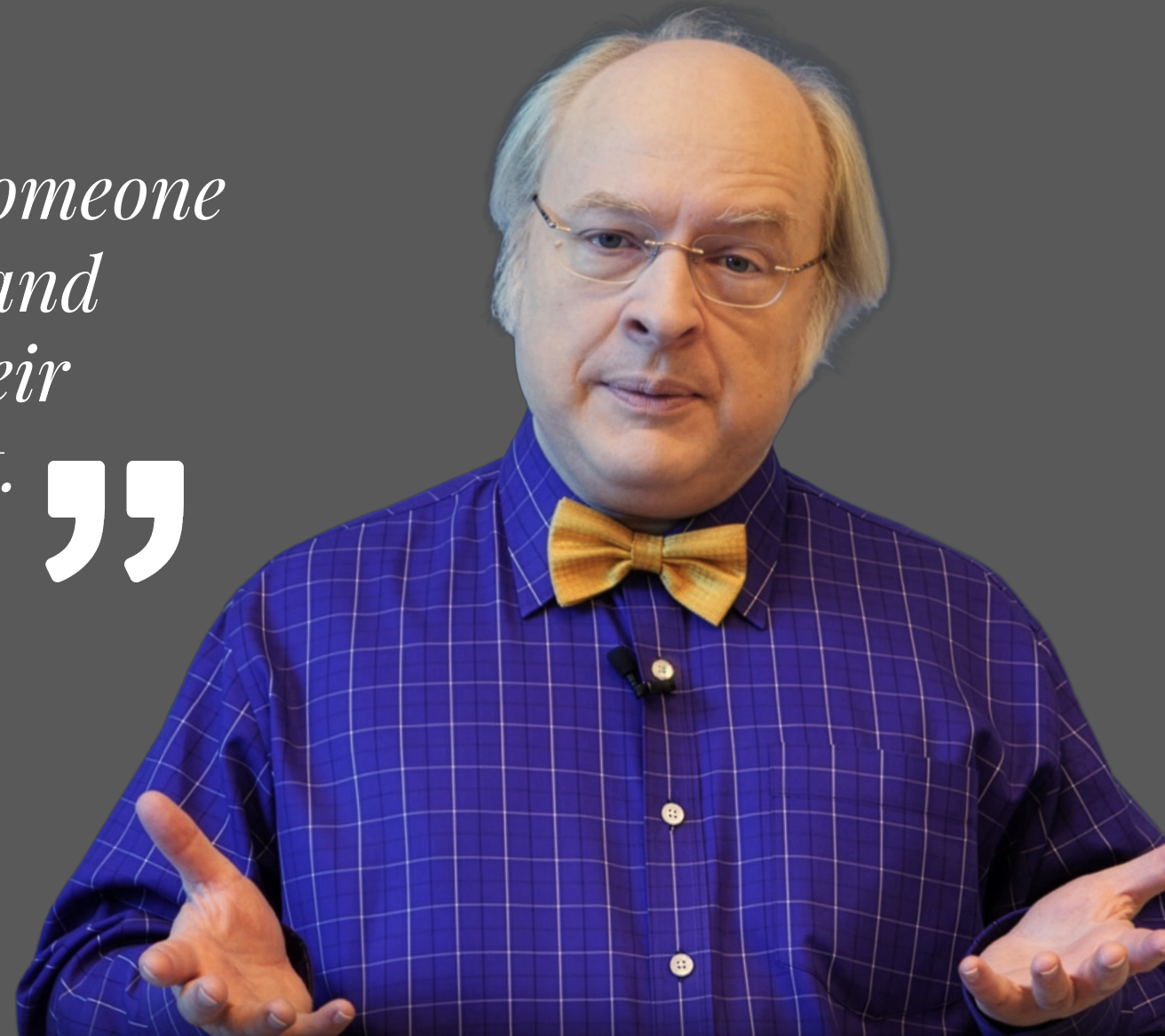
A professional toolkit to spot usability flaws

“

On average, when you ask someone to perform a task on a site, and they cannot do it. It's not their fault, it's the designers' fault.

”

- Jakob Nielsen



Usability Heuristics



Visibility of
System Status

1



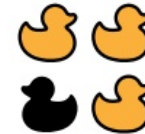
Match Between
System & Real World

2



User Control
And Freedom

3



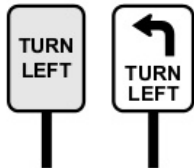
Consistency
And Standards

4



Error
Prevention

5



Recognition
Rather Than Recall

6



Flexibility And
Efficiency of Use

7



Aesthetic And
Minimalistic Design

8



Help Users
With Errors

9



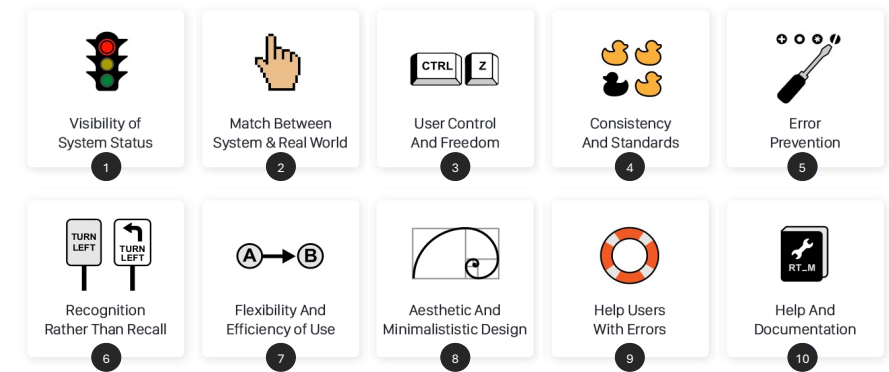
Help And
Documentation

10

Heuristic Evaluation

Expert evaluators are checking an interface for established usability heuristics

- **Group of usability experts** serving as evaluators
- They **independently** inspect a user interface against established usability principles (“heuristics”)
 - + note any issues violating the principles
 - + rate issues by severity (optional)
 - + give recommendations how to solve the issues (optional)

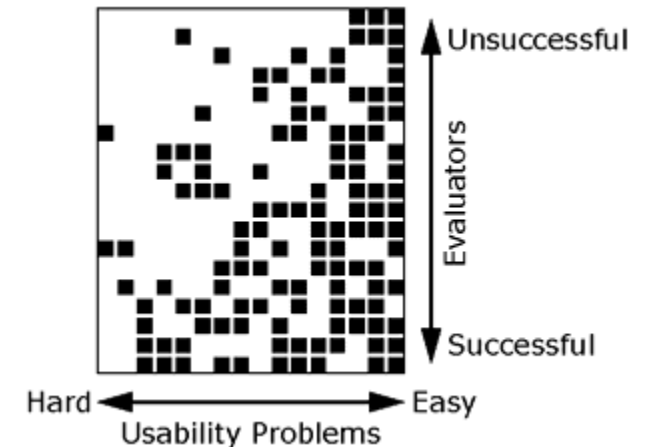


Heuristic Evaluation

Expert evaluators are checking an interface for established usability heuristics

Group of usability experts serving as evaluators

- They **independently** inspect a user interface against established usability principles (“heuristics”)
 - + note any issues violating the principles
 - + rate issues by severity (optional)
 - + give recommendations how to solve the issues (optional)
- Findings are aggregated across all evaluators, and prioritized by number of occurrences & severity



A man with short brown hair and a light beard is looking directly at the camera. He is wearing a white t-shirt. In front of him is a laptop, which is partially visible at the bottom of the frame. The background is a dark blue gradient. A large, dark grey rectangular box is overlaid on the center of the image, containing the word 'VIDEO' in large white letters and a URL below it.

VIDEO

www.youtube.com/watch?v=Y6B-u173hdQ



Usability Uncover

Usability Issues...? Let's find them all!

USABILITY UNCOVER



INPUT

usability heuristics
and examples



INSPECT

independent usability
inspection of our real-
world example: LEVIS

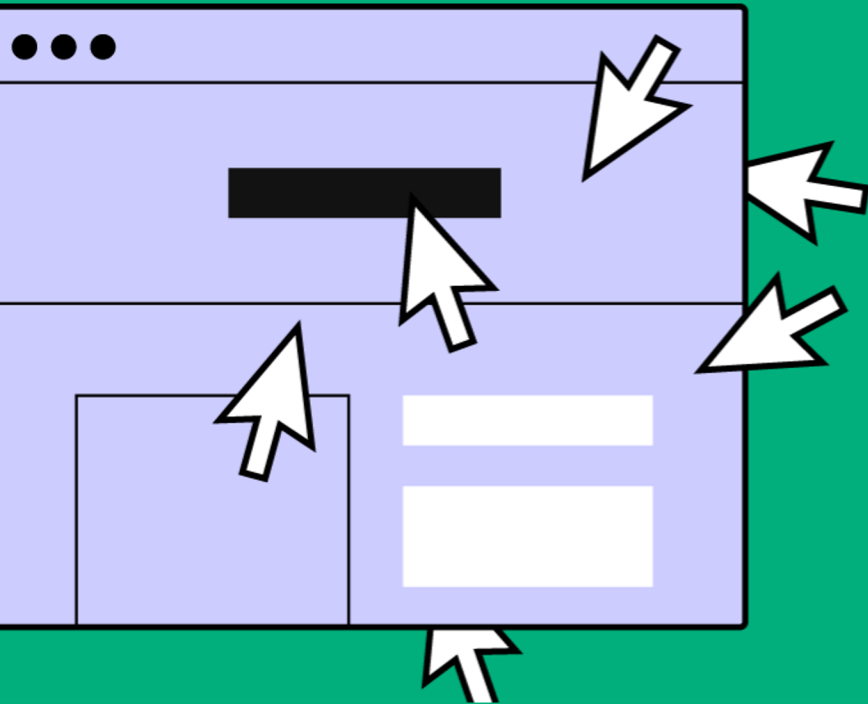


DISCUSS

consolidation &
discussion of
usability issues

Challenge #2

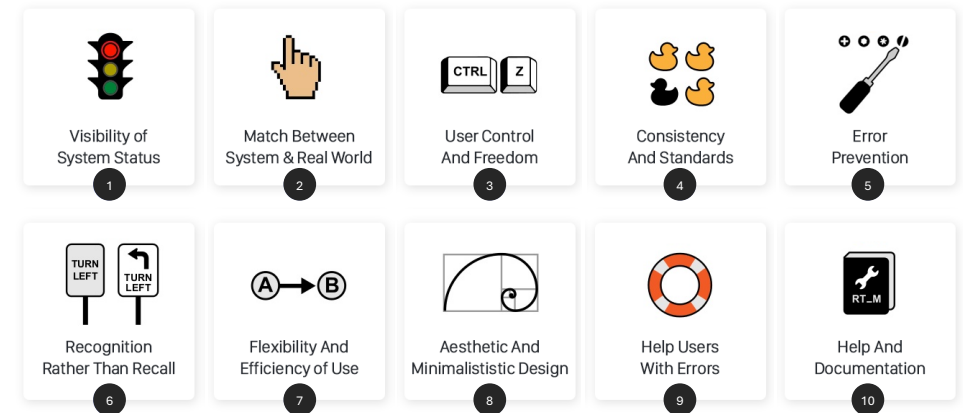
<https://elearning.fh-ooe.at/course/view.php?id=43695>



Heuristic Evaluation 101

1. Define evaluation subject and heuristics

- Heuristic Evaluation of User Interfaces* (Nielsen & Molich, 1990)
- Golden Rules of Interface Design (Shneiderman, 1987)
- Universal Principles of Design (Lidwell et al., 2003)
- First Principles of Interaction Design (Tognazzini, 2004)
- Revised Set of Usability Heuristics (Pribeanu, 2010)
- Interaction Design Principles (Cooper et al., 2010)
- SMARTphone's uSability Heuristics (Inostrosa et al., 2016)
- Usability Heuristics for Mobile (Marcos et al., 2018)
- ...



*most universally used

Heuristic Evaluation 101

1. Define evaluation subject and heuristics

2. Select evaluators

- number of evaluators
- required expertise (usability, domain-specific knowledge)

Heuristic Evaluation 101

1. Define evaluation subject and heuristics

2. Select evaluators

3. Set up a consistent evaluation system

- severity rating scale (e.g., 0-4, minor/moderate/severe, percentage, etc.)
- recommendations yes/no?

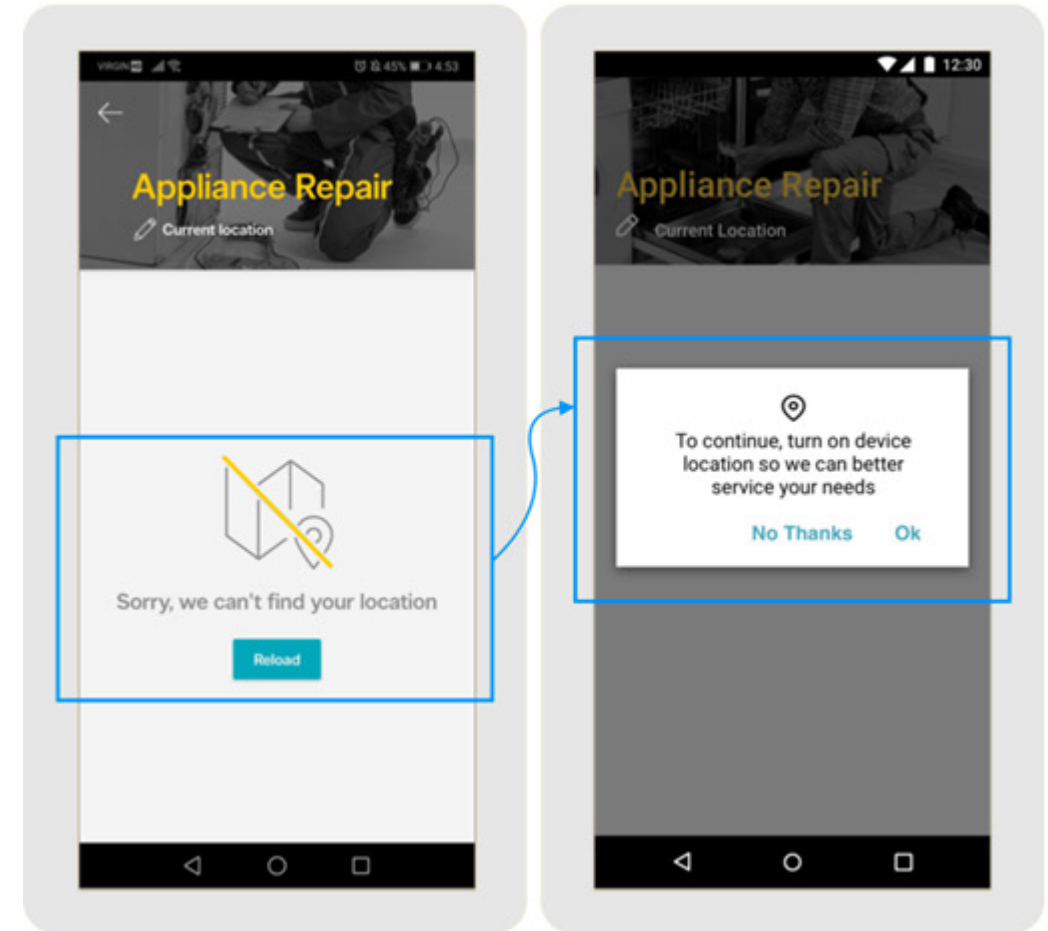
0	I don't agree that this is a usability problem at all
1	Cosmetic problem only: need not be fixed unless extra time is available on project
2	Minor usability problem: fixing this should be given low priority
3	Major usability problem: important to fix, so should be given high priority
4	Usability catastrophe: imperative to fix this before product can be released

#9 Recognize & recover from errors

In case of no location detection on the system, an error message occurs. The reload CTA has no effect when clicked + there is no information for users on how to recover / solve the error.

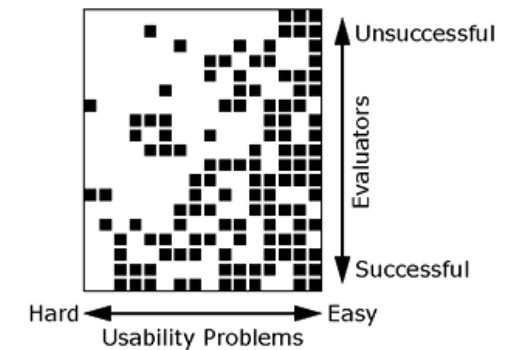
Suggestion:

Prompt user on how to actually resolve the issue (e.g., ask for permission to turn on GPS location)



Heuristic Evaluation 101

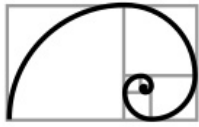
1. Define evaluation subject and heuristics
2. Select evaluators
3. Set up a consistent evaluation system
4. Evaluators walk through the interface independently, noting all violations they encounter
5. **Compare and aggregate results from all evaluators**
 - count number of occurrences for identical issues
+ overall severity (optional)
+ collect suggested improvements (optional)
 - group similar issues



Heuristic Evaluation 101

1. Define evaluation subject and heuristics
2. Select evaluators
3. Set up a consistent evaluation system
4. Evaluators walk through the interface independently, noting all violations they encounter
5. Compare and aggregate results from all evaluators
6. **Combine into prioritized list of usability issues**
7. **Derive Improvements (Redesign Suggestions)**

Usability Heuristics



Aesthetic And
Minimalistic Design

8



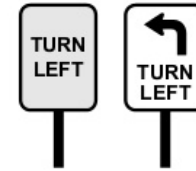
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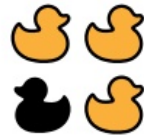
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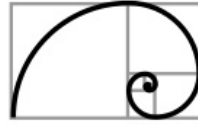
Help Users
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Aesthetic & Minimalist Design



First Impression



**MAKE A GOOD
FIRST IMPRESSION**

...YOU'VE ONLY GOT ONE.

50
milliseconds

The time that it
takes for users to form an
opinion after seeing an app
for the first time

2.6
seconds

The time that it takes for a
user's eyes to land on an area
that most influences their first
impression

94%

of first impressions are
design-related

AESTHETIC-USABILITY EFFECT.

*Users are more tolerant of minor usability issues
when they find an interface visually appealing.*



Less is more

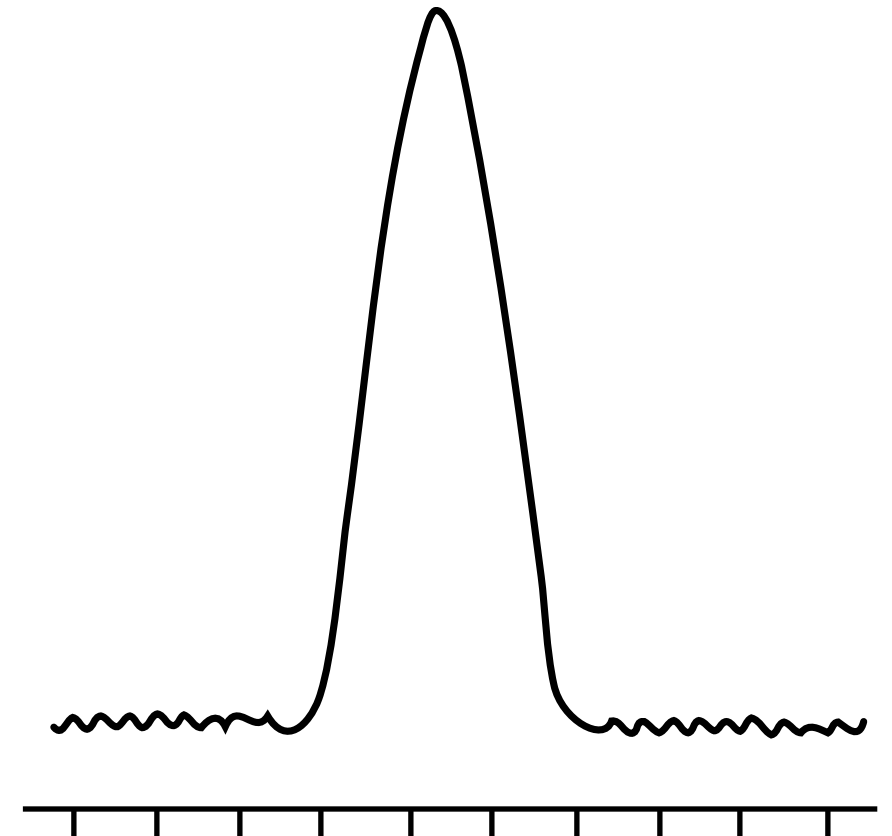
Less is More

A minimalist design limits the amount of “noise” (irrelevant information) in an interface in order to emphasize relevant information:

- 💰 every piece of information increases processing ‘cost’
- 🎯 focus on the most relevant information (user goals)
- 📄 create visual hierarchy

examples?

High Signal-to-Noise Ratio
(low noise)



[About](#) [Store](#)

[Gmail](#) [Images](#)



[Sign in](#)



Google Search

I'm Feeling Lucky

[Advertising](#) [Business](#) [How Search works](#)

[Privacy](#) [Terms](#) [Settings](#)

✓ **High SNR.**

Google website focuses on the required information to perform the primary search task. www.google.com



The counteroffensive Trump didn't seem to see coming

When Donald Trump asked for an independent review of documents taken from his Mar-a-Lago estate, he cracked open a door, allowing the Justice Department to seize the initiative.
Trump goes back on offense »



One of hardest punchers in boxing history dies at 78



'Next up, the road to recovery': Carson Daly



Ukrainian tennis player says no to handshake after match



Lawsuit alleges 2 comedians 'groomed,' 'molested' kids



'SNL' shakeup continues with 3 more reported defections

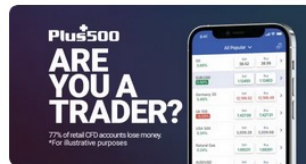
Stories for you



Business National Review

22,000 Colorado Residents Prevented from Turning Up Air Conditioning during 'Energy Emergency'

22,000 Coloradans were unable to adjust their thermostats on a sweltering summer day due to an "energy emergency."



Ad Plus500

Plus500 CFD trading Platform

77% of retail CFD accounts lose money. Trading is not for everyone, but if you are a trader - Plus500 is the place for you

Trending Now

- Cristina Kirchner
- Bruce Willis
- The Rings Of Power
- Kellye Sorelle
- Toast Payroll
- Kim Kardashian
- Chevrolet Blazer EV
- Mattress Ratings 2022
- Paulina Porizkova
- Tom Brady

Labor Day Sale

Weather

Linz

Today	Sat	Sun	Mon
72° 53°	74° 53°	76° 58°	77° 58°

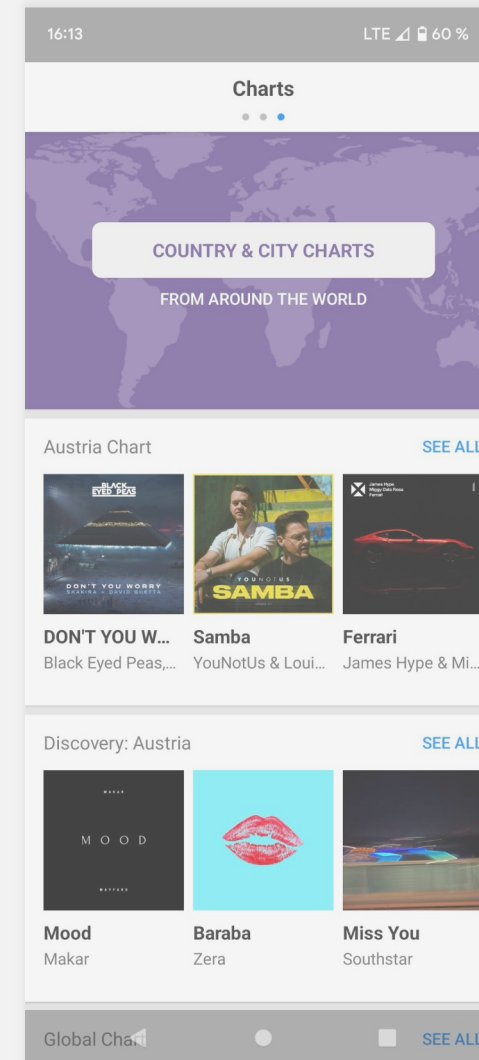
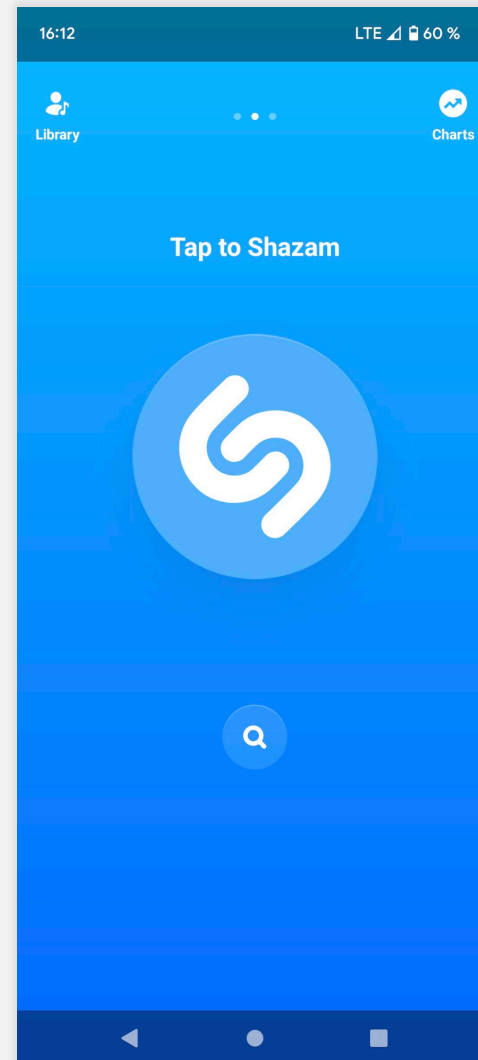
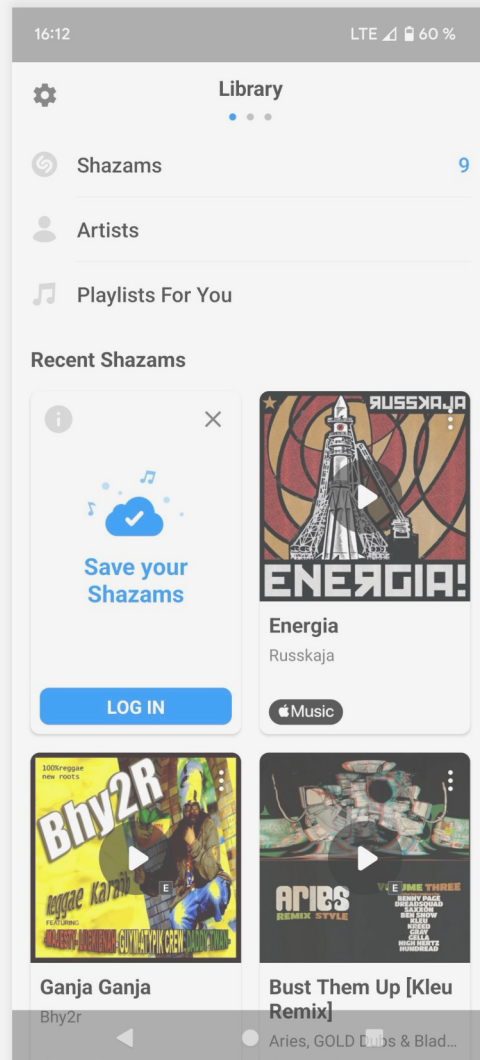
[See More »](#)

Scoreboard

Trending

Yesterday	Today	Tomorrow
Baltimore	3	Final
Cleveland	0	
Colorado	0	Final
Atlanta	3	
Milwaukee	0	Final
Arizona	5	

! Low SNR.



✓ High SNR.

Shazam app focuses on the main task, non-primary features hidden in simple navigation.

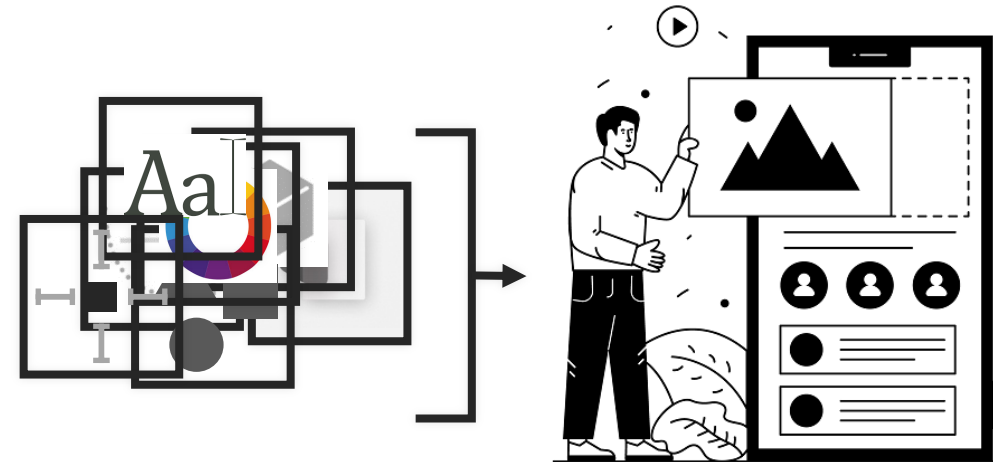


Design Systems

Design Systems

Design systems are a powerful means of creating a **shared language and visual consistency** across multiple pages and channels. They usually include elements such as style guides, component libraries, and pattern libraries.

- Style Guides
- Component Libraries
- Pattern Libraries





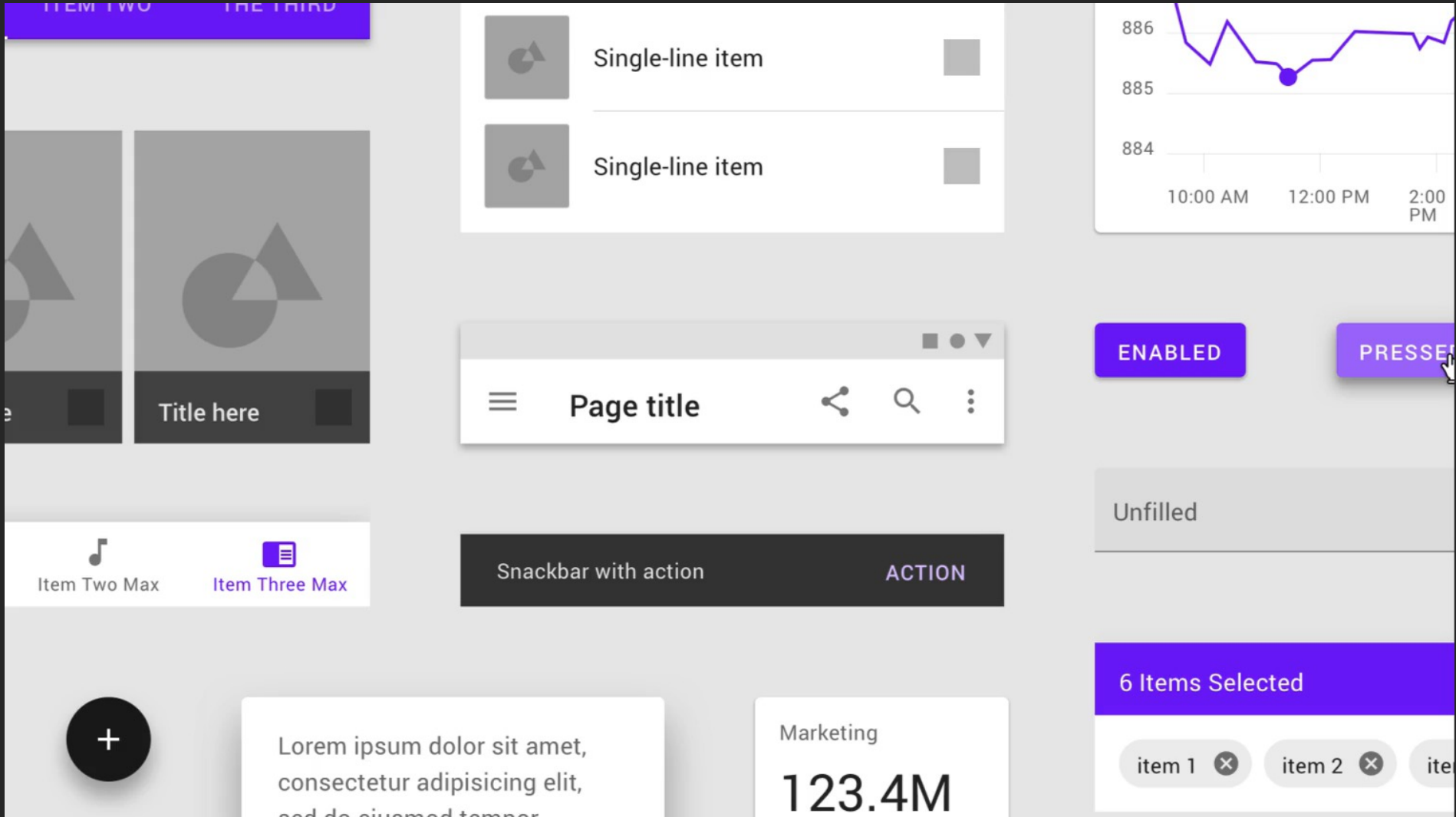
MATERIAL DESIGN

<https://m3.material.io>



Human Interface Guidelines

[developer.apple.com/design/
human-interface-guidelines](https://developer.apple.com/design/human-interface-guidelines)

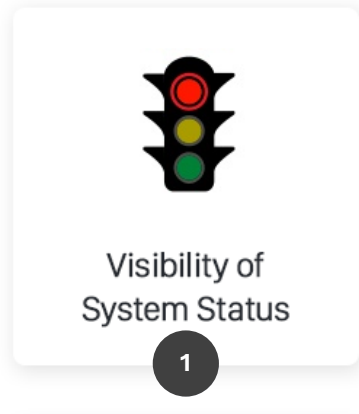




TAKEAWAY

- **First impression** is key! (mainly visual)
- Aesthetics may impact **perceived usability**
- Focus content and visual design on the **essential**
+ avoid unnecessary elements as far as possible (cognitive overload!)
- **Prioritize** content and features based on primary user goals
- Use of **Design Systems** to streamline UI

COMMUNICATE.
DON'T DECORATE.



Visibility of System Status



Navigation



Let the journey begin

☒ Roundtrip ☐ One way ☐ Multi-city

From	To	Depart	Return	Cabin Class & Travelers
Vienna (VIE)	 Country, city or airport	10/1/22	3/1/23	1 adult, Economy 

☐ Add nearby airports

☐ Add nearby airports

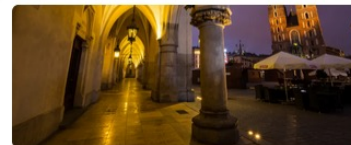
☐ Non-stop flights only

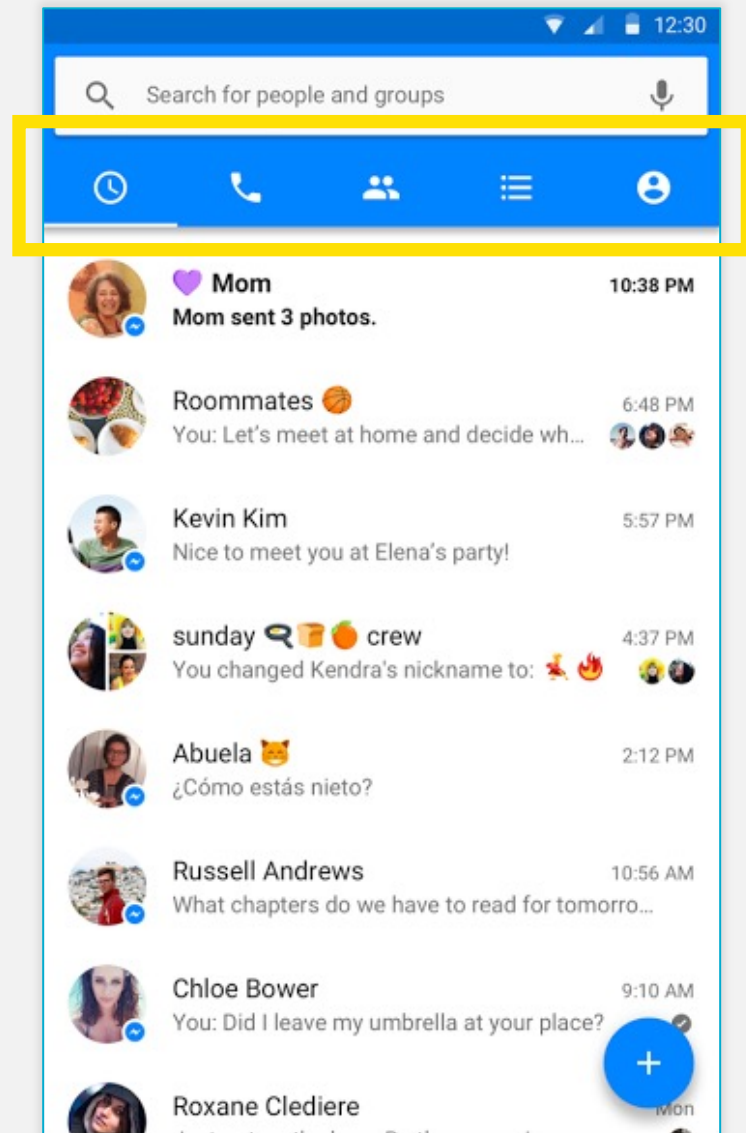
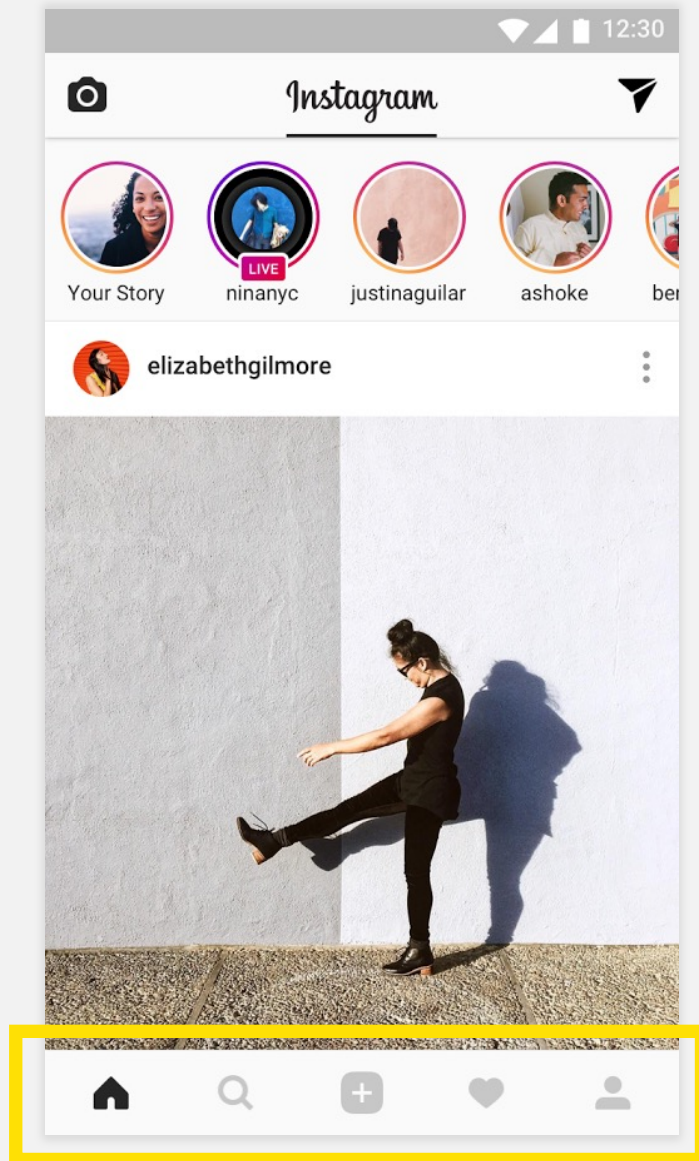
Search flights →

The best flight deals from Vienna

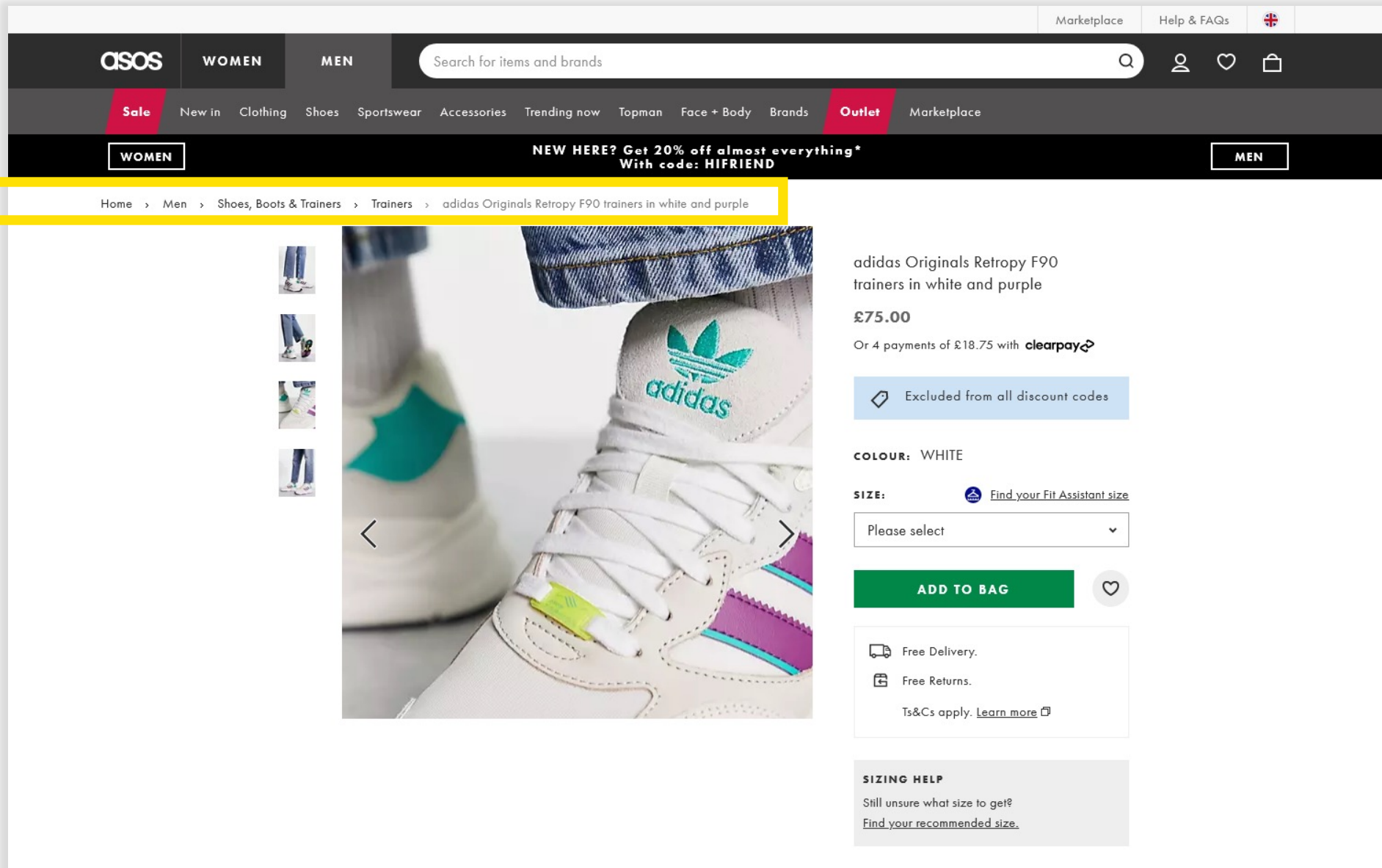
Get back out there for less with the cheapest flights we've found this week.

[See more deals](#)





other examples?



← CONTINUE SHOPPING



1 ADDRESSES

2 PAYMENT

3 REVIEW

PAYMENT METHOD

+ Pay with Gift Card

+ Add Promo Code

☒ Pay With Credit Card

Select Card

Master Card



VISA

Card Number

Expiration Date

01

/ 18

Security Code

Order Summary

[Edit Bag](#)



21313

£69.99

Ship in a Bottle

Available now

Warning!
Choking Hazard.
Small parts.



QTY1

£69.99

Subtotal £69.99

Standard shipping
(3-7 business
days) Free

Inclusive of VAT £11.67

Order Total £69.99

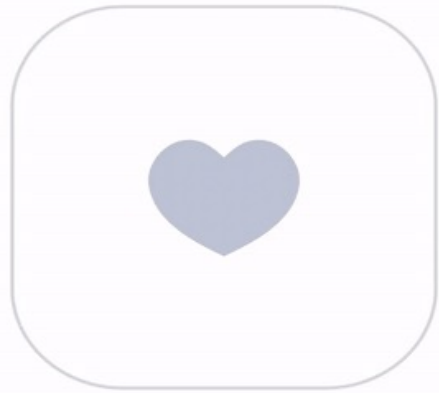


Feedback

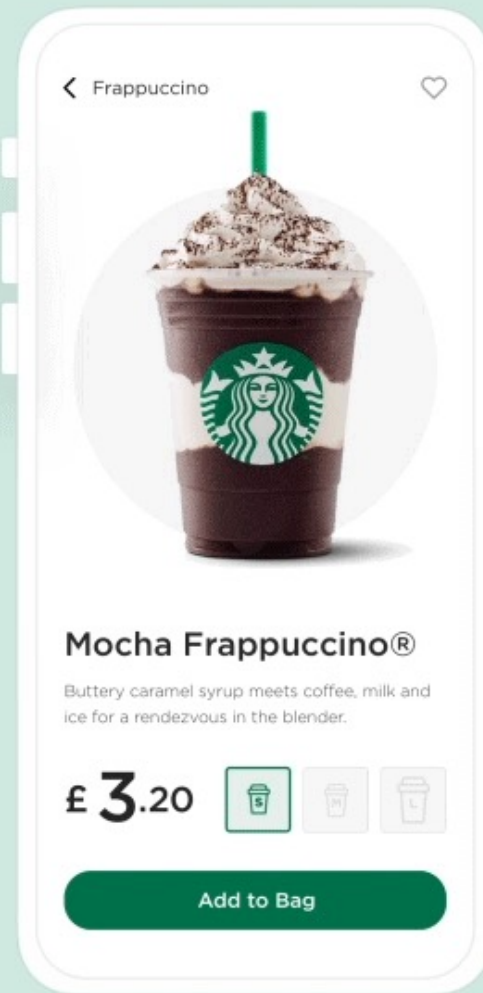


BUTTON

Visuelles Feedback dient dazu, um dem Nutzer zu versichern, dass seine/ihre Aktion von der App verstanden worden ist. In der einfachsten Form bedeutet das, dass ein Tap auf einen Button registriert worden ist. Alternative (zusätzliche) Feedback-Methoden können **vibrotaktilen oder auditiven Feedback** sein.

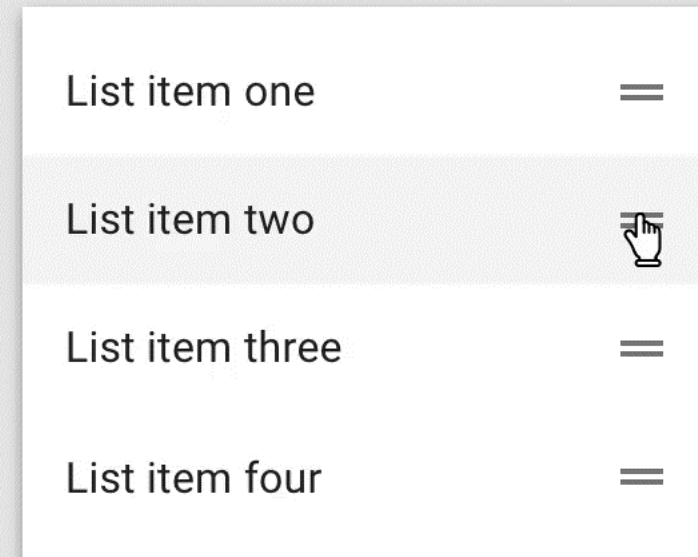
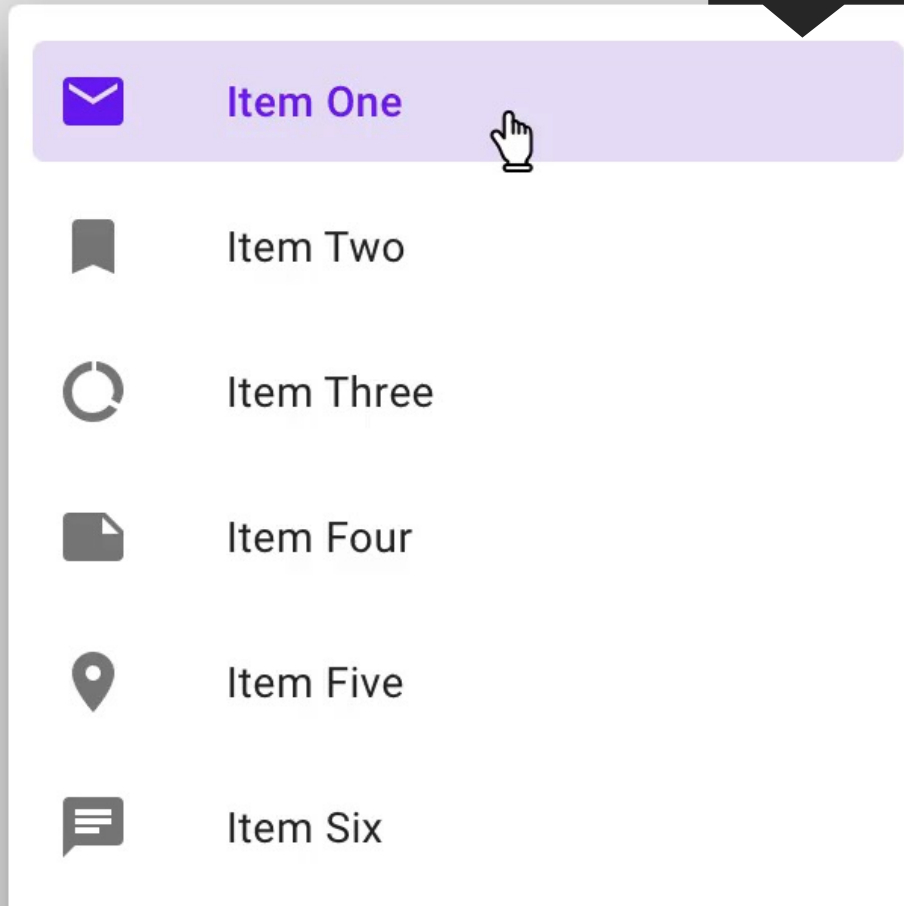


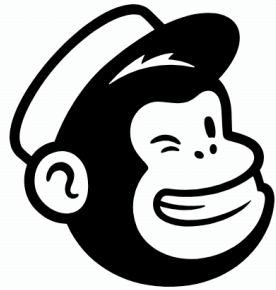
In manchen Fällen ist es auch wichtig, den **Status** eines UI-Elements zu verändern. In diesem Fall kommuniziert das Feedback zusätzlich das Ergebnis einer Interaktion. Dadurch wird die Aktion nicht nur **sichtbar**, sondern auch **nachvollziehbar**.



other
examples?

other
examples?





Get started with a free account

Find your people. Engage your customers. Build your brand. Do it all with Mailchimp's marketing tools. Already have an account? [Log in](#)

Email

Username

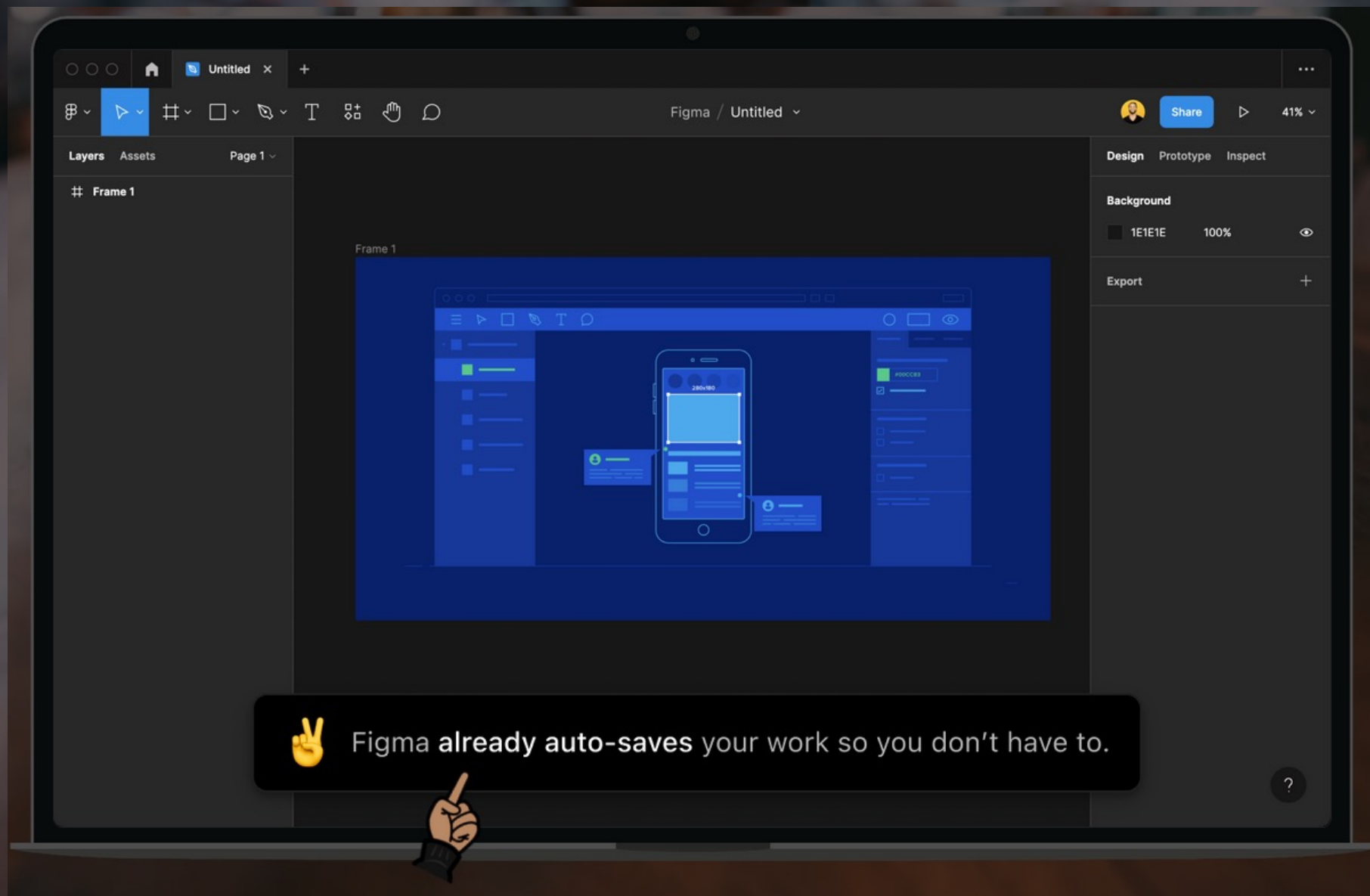
Password

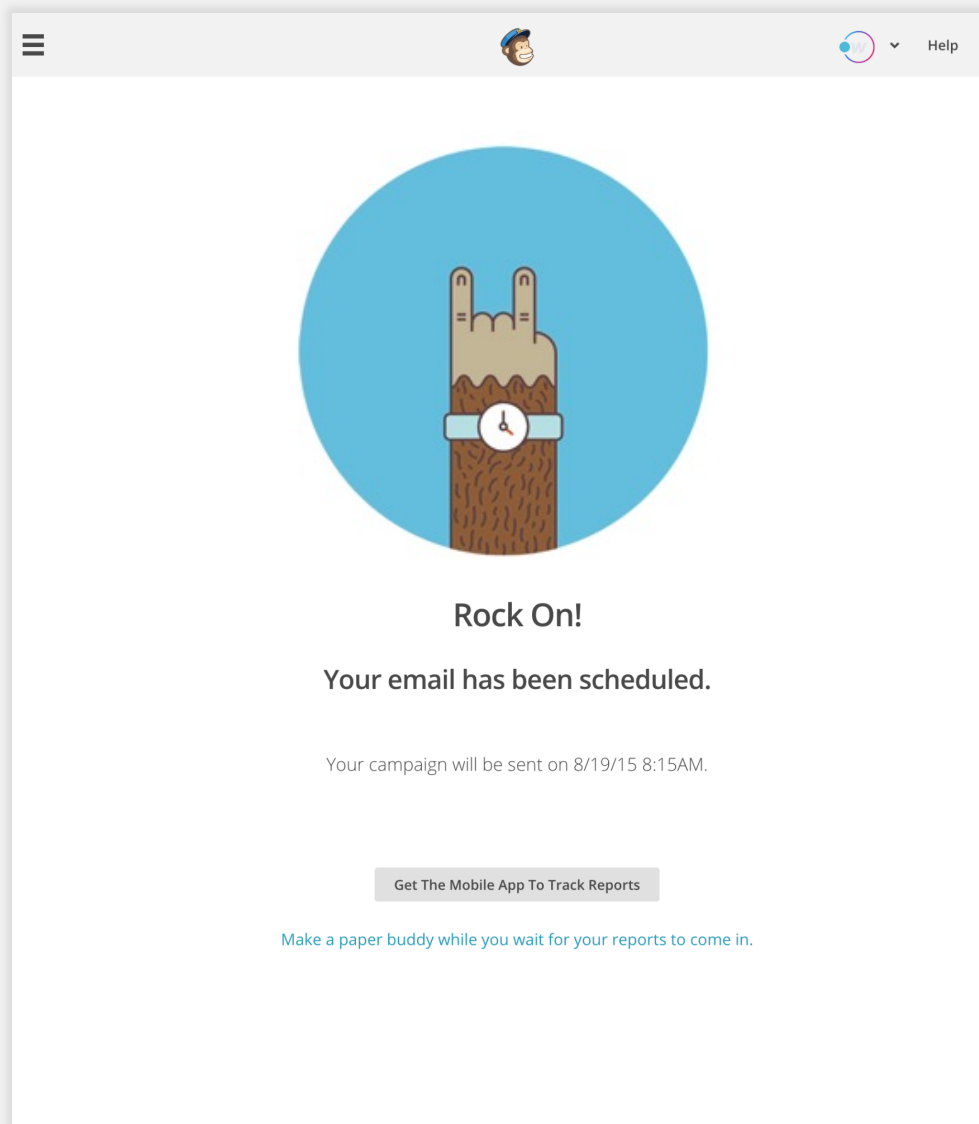
 [Show](#)

- One lowercase character
- One uppercase character
- One number
- One special character
- 8 characters minimum

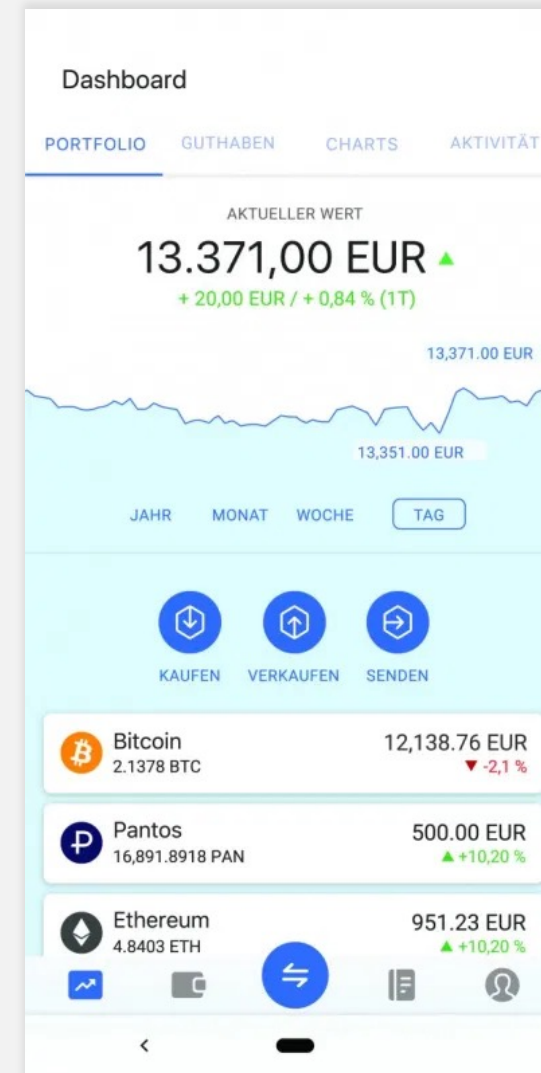
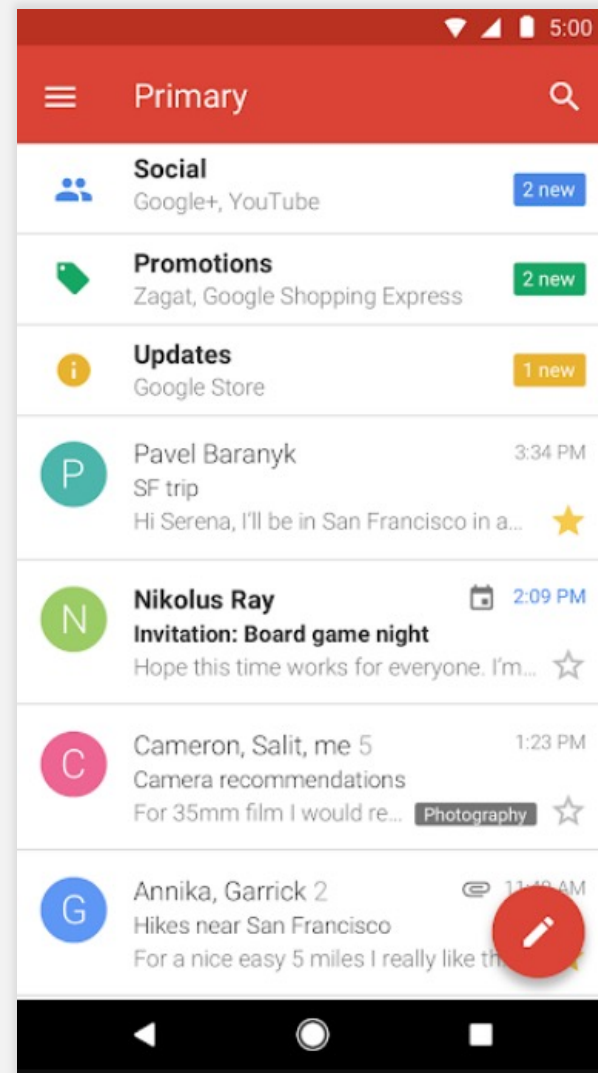
Get Started!

By clicking this button, you agree to Mailchimp's [Anti-spam Policy & Terms of Use](#).

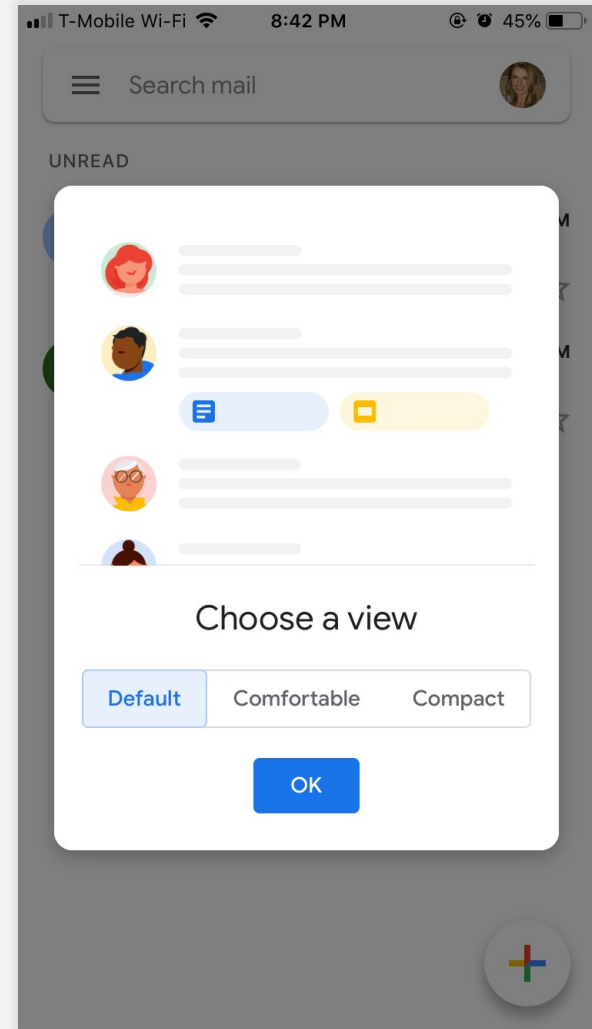
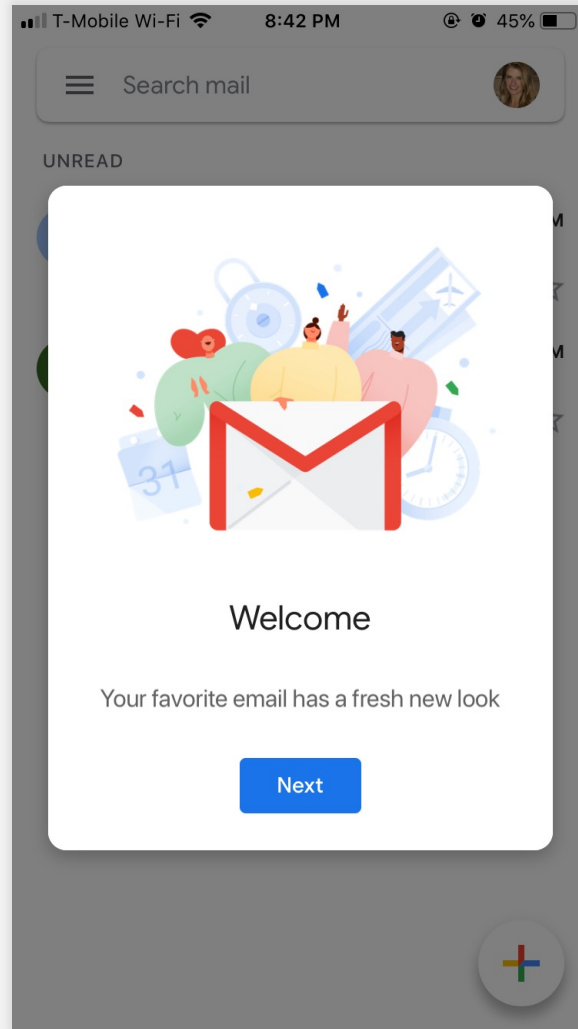
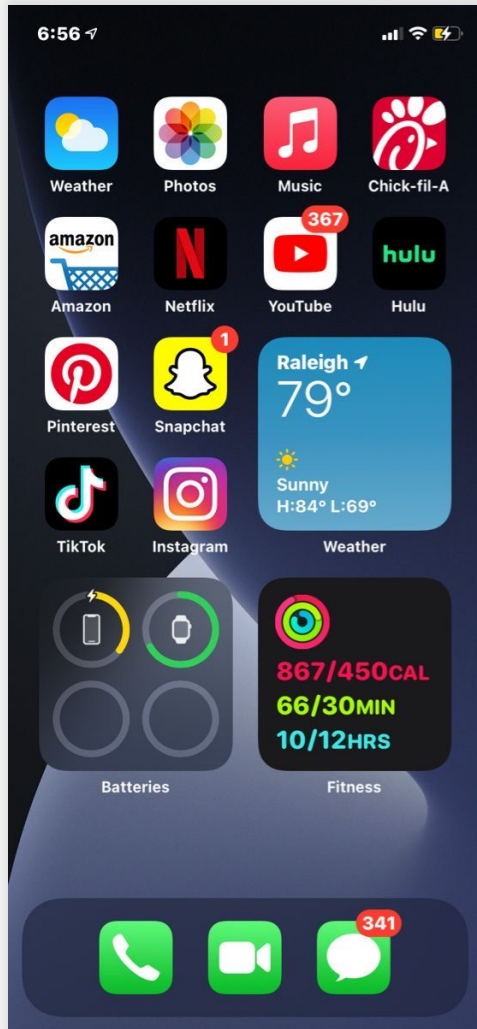




Mailchimp confirmation screen after email campaign has been scheduled, dribbble.com/shots/1550568-MailChimp-Rock-On



examples for indicators in Gmail and Bitpanda apps



examples for notifications on iOS home screen (passive), and in Gmail app (active)



Response Times

Response Times

**0.1
Seconds**

is perceived as
immediate

>> limit for having the user
feel that the system is
reacting instantaneously

**1
Second**

is perceived as
noticeable delay

>> limit for the user's flow
of thought to stay
uninterrupted

**10
Seconds**

is perceived as
waiting time

>> limit for keeping the user's
attention focused on the
dialogue

Progress Indicators

Use a progress indicator for any action that takes longer than about 1 second:

- **Confirmation** for users that the system has received their request
- Reduces **perceived waiting time**
- Indicates **expected duration** (e.g. „x% done“)



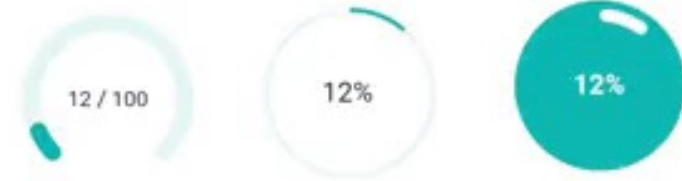
examples?

In general, progress indicators should be used for delays >1 seconds



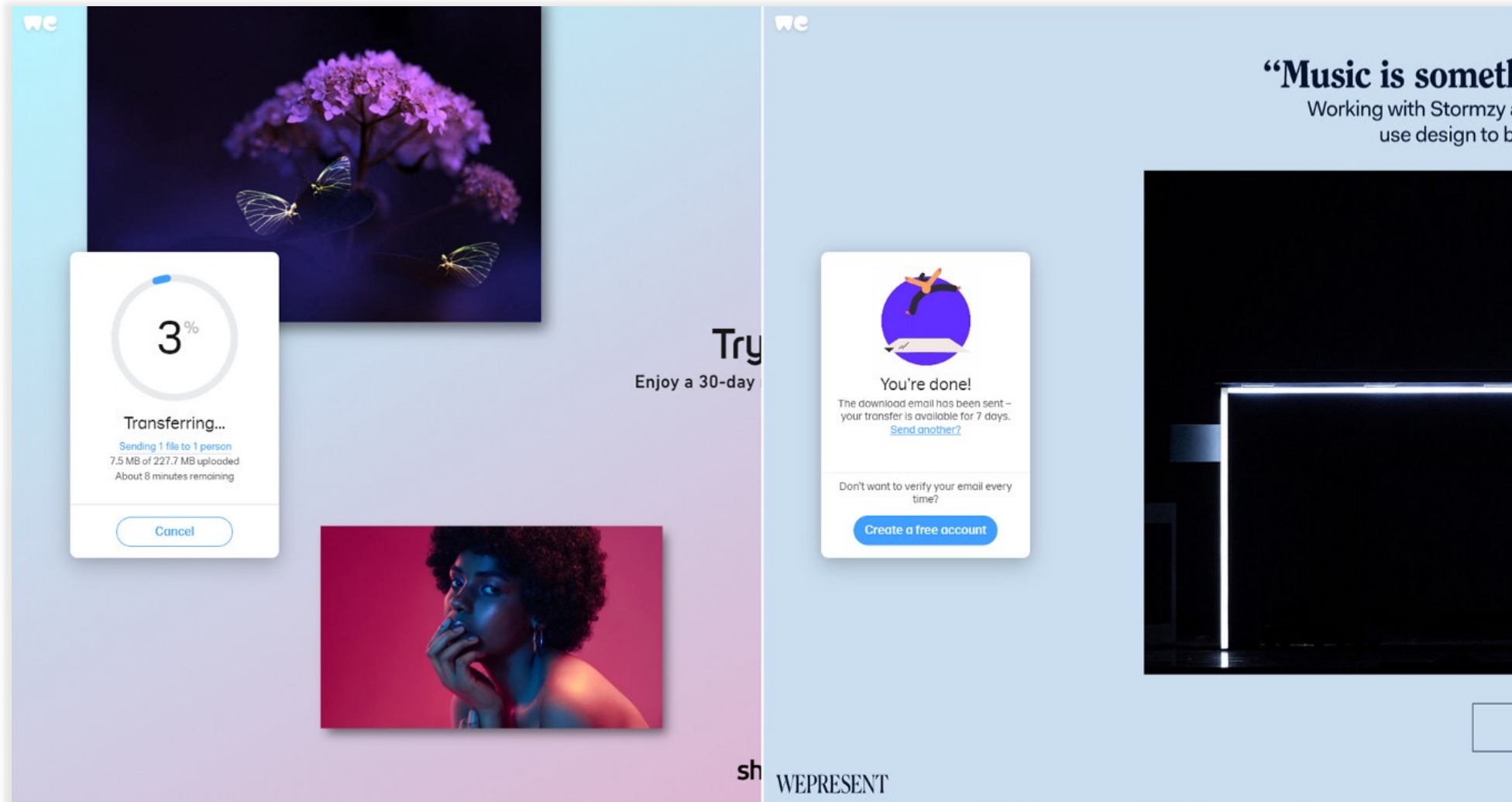
Looped Animation (1-10 seconds)

offers feedback that the system is working,
but it doesn't give any information about how
long the user will have to wait

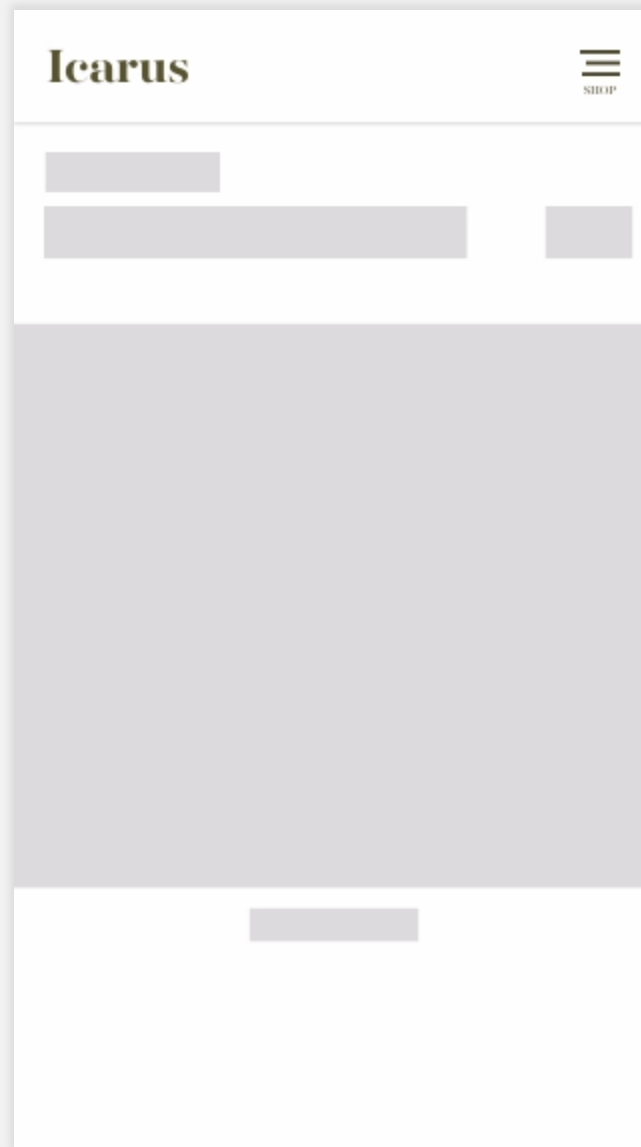


Percent-Done Animation (> 10 seconds)

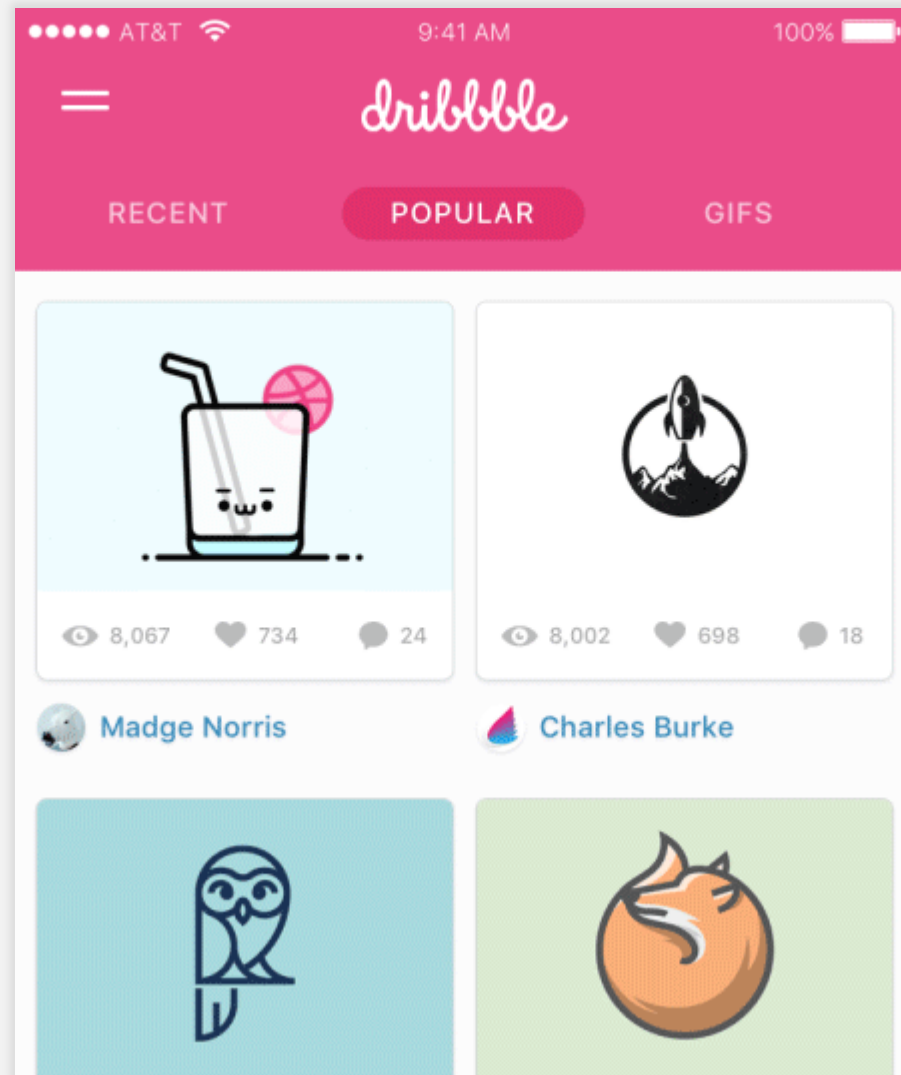
show the current progress, how much has already
been accomplished, and how much is left
(approximation of waiting time)



WeTransfer progress bar communicating upload state, uxplanet.org/1-visibility-of-system-status-simplified-by-the-examples-1cca76f59b6a



special case: skeleton screens bridging waiting times for async content fetching, uxdesign.cc/what-you-should-know-about-skeleton-screens-a820c45a571a



special case: pull-to-refresh animations providing feedback on content updates in mobile interfaces, dribbble.com/shots/2590603-Liquid-Pull-Down

TAKEAWAY

- Provide **feedback as fast as possible** (functional, visual / auditory / tactile)
+ use loading animations to bridge waiting times
- Communication of system state creates **transparency and predictability**,
which results in trust and feeling of control for users
- Up-to-date information creates **overview and awareness** (discoverability!)

KEEP USERS INFORMED
THROUGH APPROPRIATE
FEEDBACK WITHIN
REASONABLE TIME

Usability Heuristics



Aesthetic And
Minimalistic Design

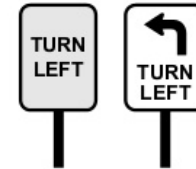


Visibility of
System Status



Match Between
System & Real World

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Recognition
Rather Than Recall

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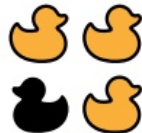
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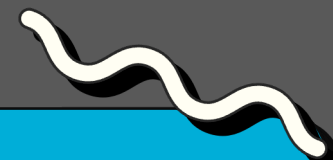
Help Users
With Errors

9



Help And
Documentation

10



Interface Design

IDE2IL

up next: IDE2IL 25.03.



kathrin.probst@fh-ooe.at





Interface Design

IDE2IL



kathrin.probst@fh-ooe.at

